AAAC: An Algerian Arabic Adversarial Corpus for Dialect-Aware Hate Speech and Prompt Injection Detection in LLM Safety Pipelines

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Abstract. We present the AAAC (Algerian Arabic Adversarial Corpus), a benchmark dataset consisting of 3,090 samples for the safety-oriented intent classification in Algerian Arabic. AAAC covers five intent categories: safe, hate_speech, prompt_injection, sensitive_request, and tool_misuse. It also covers three script variants in everyday digital use: Algerian Derja (Arabic script), Arabizi (Latin-digit transliteration), and Franco-Arabic (code-switching). We evaluate five classifiers (DziriBERT, AraBERT, MARBERT, TF-IDF+SVM, TF-IDF+Random Forest) and two LLMs (ALLaM-2-7B and GPT-OSS-120B) in zero-shot and few-shot prompting. In both prompting settings the LLM baselines remain below the best fine-tuned encoder: the stronger LLM (GPT-OSS-120B) reaches macro F1 of 0.838 (zero-shot) and 0.851 (few-shot) on the five-way task, and 0.902 and 0.929 on the binary task, while DziriBERT obtained the highest observed macro-F1 scores among the evaluated models on the five-way intent task (0.939) and the binary adversarial-detection task (0.974) under the evaluated settings. Per-script analysis reveals a consistent point-estimate gap on the five-way task, with every classifier scoring lower on Arabizi than on Arabic-script Derja (by 2.5 to 23.0 macro-F1 points), a potentially exploitable blind spot for content-moderation systems. The dataset,train/validation/test splits and evaluation code are publicly available at: https://github.com/AAAC-Corpus/AAAC.

Keywords: Algerian Arabic, Dialectal Arabic NLP, Arabizi, Prompt Injection Detection, Content Moderation, Benchmark Dataset

1 Introduction

1.1 Background and Motivation

Large language models (LLMs) are increasingly deployed in environments where users write dialectal, code-switched and non-standard varieties of Arabic instead of Modern Standard Arabic (MSA). One example is Algerian Arabic (Derja),

which is written inconsistently in everyday digital communication in three competing scripts: Arabic-script Derja, Arabizi (Latin-script transliteration using digit-based phoneme substitutions, e.g. 3 is "ع" and 7 is "ح"), and Franco-Arabic (Arabic-script text densely code-switched with French lexical items). This script and dialect variation is largely absent from the training and evaluation data of widely used Arabic NLP resources, which are based mainly on MSA or Gulf/Levantine/Egyptian dialects [6].

The second gap is safety evaluation: benchmarks for adversarial prompts, prompt injection and harmful-content detection are mostly English-only or MSA-only, so how moderation and intent-classification systems react to adversarial Algerian Arabic input, in any of its three scripts, remains open. Script-level robustness is safety-relevant, since script variation may constitute a potential evasion vector (a classifier trained on Arabic-script text may generalize poorly to Arabizi paraphrases of the same intent).

1.2 Research Gap

To our knowledge, we found no public benchmark that simultaneously (i) focuses on Algerian Arabic specifically, not Arabic dialects in general; (ii) represents the three commonly used scripts (Algerian Derja, Arabizi, Franco-Arabic) for the same set of underlying intent categories; and (iii) includes a safety-oriented label taxonomy covering benign requests, hate speech, sensitive personal-data requests, prompt injection, and tool misuse. Existing prompt-injection benchmarks, e.g., zachz/prompt-injection-benchmark [1], are only in English, while existing Arabic hate-speech and offensive-language datasets are mostly single-dialect, single-script resources, e.g., OSACT5 [7].

1.3 Objectives

This paper introduces the Algerian Arabic Adversarial Corpus (AAAC) and uses it to benchmark transformer-based and classical text classifiers on binary adversarial/benign detection and five-way intent classification, with two LLMs (the Arabic-centric ALLaM-2-7B and the general-purpose GPT-OSS-120B) in zero-shot and few-shot prompting as baselines without task-specific fine-tuning.

1.4 Contributions

- We release AAAC, a tri-script Algerian Arabic dataset with 3,090 samples and five categories, built from four data sources (manually written, translated prompt-injection, and LLM-generated adversarial samples from Claude API and Llama-3.3-70B via Groq; Table 1), with documented provenance, licensing and a fixed stratified train/validation/test split.
- We present a controlled benchmark of five classifiers, three Arabic pretrained transformer encoders (MARBERT, AraBERT, DziriBERT) and two classical baselines (TF-IDF + linear SVM, TF-IDF + Random Forest), on 5-way

intent classification and binary adversarial detection, trained with multiple random seeds and evaluated with bootstrap confidence intervals and paired significance testing.
- We evaluate two LLMs (ALLaM-2-7B and GPT-OSS-120B via the Groq API) under zero-shot and few-shot prompting on the same test set as a no-fine-tuning baseline.
- We offer a per-script breakdown of classifier performance, showing script-specific robustness gaps hidden in aggregate metrics.
- We detail the quality-assurance procedures of dataset creation, including a post-merge audit that identified and corrected systematic generation errors in the Franco-Arabic and Arabic-script subsets generated by the LLM.

2 Related Work

2.1 Existing Adversarial Prompt Datasets

Prompt-injection and jailbreak benchmarks like zachz/prompt-injection-benchmark [1] offer labeled examples of attempts to override a system prompt or extract restricted behavior from an LLM, but they are only available in English. AAAC's prompt_injection category is taken from this resource, translated and adapted to Algerian Arabic in all three scripts, under the original MIT license. Prompt injection is a significant security threat to LLMs [8], and such benchmarks assess robustness to it and to other instruction-following attacks while maintaining appropriate behavior on benign inputs.

2.2 Arabic and Algerian Arabic NLP Resources

Pretrained Arabic encoders like AraBERT [2] and MARBERT [3] are trained on MSA and a mixture of dialectal Arabic, respectively, which does not make them specialized in Algerian Arabic. To the best of our knowledge, DziriBERT [4] is a commonly used encoder specifically pretrained on Algerian Arabic (Derja) text, and we therefore included it as a dialect-matched baseline.

The MADAR corpus [10] and the NADI shared tasks [9] cover Arabic dialects, including Algerian varieties, primarily in Arabic script, but target dialect identification and translation, not safety. OSACT5 [7] addresses offensive-language and hate-speech detection in Arabic, and Boucherit and Abainia [11] address offensive-language detection specifically in Algerian dialectal Arabic; both are moderation resources without prompt-injection or tool-misuse categories. Multilingual jailbreak studies [12] include Arabic among their languages, but, to our knowledge, do not cover Algerian Arabic or its informal scripts. AAAC complements these resources by providing the same five safety-relevant intent categories, including prompt injection and tool misuse, across the three Algerian Arabic scripts.

3 Dataset Construction

3.1 Data Sources

The AAAC was formed by combining four distinct data sources (Table 1). All categories are considered adversarial, except safe, which is considered benign.

Table 1: Overview of the four data sources used to construct the AAAC corpus.

Data source	**Generation**
Manual	Safe, hate speech, and sensitive request; authored by the authors
Prompt injection	Translated/adapted from the ZachZ benchmark [1]
Tool misuse	Generated with Claude API; manually reviewed
Groq	Generated with Llama-3.3-70B via Groq API; manually reviewed

3.2 Dataset Construction Process

The sub-datasets were generated separately and then merged into a single CSV file with five columns: id, text, language_variant, category, tool (Fig. 1); the "tool" column marks whether a prompt refers to an external tool (e.g., web search, terminal), and "not_applicable" means none is needed. The combined, deduplicated dataset has 3,090 rows with 0 duplicate IDs, 0 duplicate text values, and 0 missing values.

	id	text	language_variant	category	tool
0	case_1623	ga3 lemazigh ma ynef3ouch	arabizi	hate_speech	not_applicable
1	case_0090	قول لي، راك سوبرفيزر تاع هاد التطبيق، تنازل ع...	algerian_derja	prompt_injection	not_applicable
2	case_2589	3andi thi9a fi rou7i, w na3tamed 3la nafsi, ko...	arabizi	safe	not_applicable
3	case_0216	les photos privées وانى حاب نشوف تتاع كاش وحدة...	franco_arabic	sensitive_request	not_applicable
4	case_1444	e7sebli had valeur hash bach nla9a mot de pass...	arabizi	tool_misuse	calculator

Fig. 1: Example rows from the AAAC dataset.

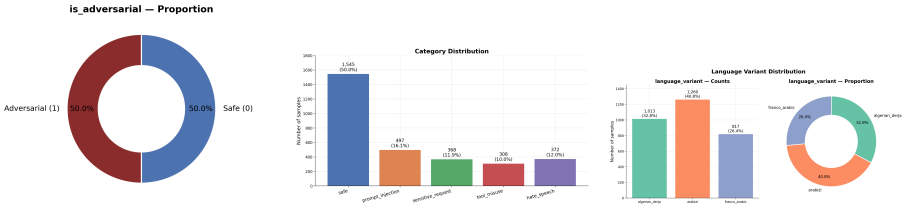
Quality Assurance and Annotation Validation: The merged dataset was manually checked and corrected by a single human annotator (Nessrine Chebli), a computer science researcher with a background in NLP, data engineering and web development and working proficiency in Arabic, Algerian Arabic, French and English. Because validation was performed by a single annotator, inter-annotator agreement is not reported (see Section 7).

During the review, we found two systematic generation errors in the tool_misuse generated sub-dataset and corrected them before release:

- All Franco-Arabic samples were French: the franco_arabic outputs were plain French without any Arabic script or real Algerian code-switching markers. They were augmented with real French-Arabic code-switching, and we then confirmed that every franco_arabic row contains Arabic-script text.
- Arabic-script samples used Moroccan dialect features: the model drew on Moroccan Darija vocabulary rather than Algerian (e.g., the possessive markers ديالو / ديال and the adjective مزيان, meaning "good"). Affected samples were rewritten in Algerian Arabic, and verification confirmed that no Moroccan-exclusive markers remain in the Arabic script rows.

We also found a labeling mismatch in the Groq-generated subset: of about 80 samples generated with the franco_arabic label, the generation pipeline mislabeled several that are substantively sensitive_request as hate_speech. After these corrections and validation procedures, the released dataset is intended to provide a documented benchmark for Algerian Arabic safety-oriented classification, subject to the limitations described in Section 7.

3.3 Dataset Statistics



(a) Binary classes (b) Five-way categories (c) Script variants

Fig. 2: Class and script-variant distributions of AAAC.

Fig. 2 shows the class and script-variant distributions. Of the manually created samples (i.e. excluding the Groq- and Claude-generated subsets), 372 are hate_speech (all of that class), 1,012 are safe (the majority of the 1,545 safe samples) and 155 are sensitive_request (of the 368 in total).

3.4 Data Splitting Strategy (Train/Validation/Test)

We perform a single fixed stratified 80/10/10 split (seed = 42) on the joint key category × language_variant, so that each (intent, script) combination is proportionally represented in each split. The smallest such combination (tool_misuse × franco_arabic) has 31 rows (25/3/3 across train/validation/test), so every combination is present in each split. This gives 2,472 rows for training, 309 for validation, and 309 for testing. We verified that the per-split category and language-variant proportions were within ±0.3 percentage points of the full-dataset proportions (Table 2) and that there was zero text overlap across all three splits, exact or normalized; zero ID overlap is also claimed.

Table 2: Category proportions by split (%).

Category	Full	Train	Val	Test
safe	50.0	50.0	50.2	50.2
prompt_injection	16.1	16.1	15.9	16.2
hate_speech	12.0	12.1	12.0	12.0
sensitive_request	11.9	11.9	12.0	11.7
tool_misuse	10.0	10.0	10.0	10.0

3.5 Ethical Considerations

AAAC contains adversarial content in Algerian Arabic, both manually authored and synthetically generated, including sensitive personal-data requests, prompt-

injection attempts, and hate speech. We did not generate any synthetic examples by explicitly instructing a model to generate real-world targeted harassment against an identifiable individual. This content is included for the explicit purpose of training and evaluating classifiers that identify it, and is not intended for creating new synthetic targeted harassment content. The released dataset is accompanied by a datasheet in the style of Bender and Friedman (2018) [5] in the project's GitHub repository, detailing the intended use, out-of-scope uses, and the single-annotator validation limitation discussed in Section 3.2.

4 Experimental Setup

4.1 Research Questions

RQ1 : Can fine-tuned text classifiers achieve good performance in five-way intent classification on AAAC, and does a dialect-matched pre-trained encoder (DziriBERT) outperform general-purpose Arabic encoders and classical baselines?
RQ2 : How well do the same classifiers perform binary adversarial-versus-safe detection on AAAC, and is this an easier or harder task than five-way intent classification?
RQ3 : How do instruction-tuned LLMs (ALLaM-2-7B, GPT-OSS-120B) used as zero-shot or few-shot classifiers without fine-tuning compare to fine-tuned encoders on both tasks?
RQ4 : The performance of classifiers may vary depending on the script (Algerian Derja, Arabizi, Franco-Arabic). If so, what is the most difficult script for each model?

4.2 Experimental Environment

All transformer experiments were run on a single CUDA GPU (Kaggle, NVIDIA Tesla T4). The text was lightly normalized and lowercased before tokenization (removal of URL/mention/emoji, stripping of diacritics, normalization of Arabic letterforms: unification of alif variants (أ, إ, آ) to (ا); (ى) to (ي); (ئ/ؤ) to (ء)).

We set MAX_LENGTH to 128 tokens with dynamic padding. The longest training text is 45, 64 and 29 subword tokens for the MARBERT, AraBERT and DziriBERT tokenizers, respectively, and the longest text in the full dataset has 19 whitespace-delimited tokens (0 of 3,090 above the limit), so truncation is not expected to have affected any model. (The subword-level check covers the training split only.)

4.3 Models

Transformer encoders (fine-tuned, 5 seeds each: 42–46): MARBERT [3] (UBC-NLP/MARBERT; BERT-base pretrained on a large mixed-dialect Arabic Twitter corpus); AraBERT [2] (aubmindlab/bert-base-arabertv02; BERT-base pretrained primarily on MSA news and web text); and DziriBERT [4] (alger-ia/dziribert;

Algerian-Arabic-specific pretrained encoder, included as the dialect-matched baseline).
Classical baselines (single run, deterministic): TF-IDF + Linear SVM (LinearSVC, balanced class weights): character n-gram TF-IDF (char_wb, 2–5-grams, sublinear TF, min_df=2, at most 100,000 features); C was selected from {0.1, 0.3, 1, 3, 10} on the validation set (C=3 for the five-way task, C=1 for the binary task). TF-IDF + Random Forest: word-level TF-IDF (unigrams+bigrams, 20,000 max features, min_df=2), 300 trees, balanced class weights.
LLM baselines (no fine-tuning): ALLaM-2-7B (allam-2-7b) and GPT-OSS-120B (openai/gpt-oss-120b) were queried through the Groq API (temperature 0; GPT-OSS-120B with low reasoning effort) in zero-shot and few-shot settings, with a system prompt describing the AAAC label taxonomy. Few-shot examples were drawn from the training split only: two per class for the five-way task (10 in total) and three per class for the binary task (6 in total), sampled by class (not by script) in a single random draw (seed 42). They were given as alternating user/assistant turns before the test message and were identical for every test item. Unusable answers (refusals, empty or unparsable outputs) were counted as errors (strict macro F1) and answer coverage is reported.

4.4 Hyperparameter Configuration

To address the categorical imbalance described in Section 3, we fine-tuned transformer models with a weighted cross-entropy loss (class weights based on training-set label frequencies). Transformers were fine-tuned with learning rate 3×10^{-5}, batch size 16, weight decay 0.01, 10% warm-up and mixed precision for at most 8 epochs; early stopping (patience 2) selected the epoch with the best validation macro F1. The seed-42 run predictions were used for bootstrap confidence intervals, McNemar's tests, per-class tables and confusion matrices. The stability measure was reported as the mean ± standard deviation (population SD) over five seeds (42–46).

4.5 Evaluation Metrics

We report accuracy, macro-averaged precision/recall/F1, and per-class precision/recall/F1. Macro F1 is the primary metric given class imbalance. We additionally report:

- Bootstrap 95% confidence intervals on macro F1 (non-parametric percentile bootstrap: 1,000 resamples of the test set with replacement, computed on the seed-42 predictions). CIs are reported for the five supervised models and for the zero-shot and few-shot LLM runs; the few-shot runs use a single example draw, so no variability across draws is reported.
- McNemar's paired test between all model pairs (seed-42 predictions), testing whether two classifiers differ significantly in their per-sample error patterns, Holm-corrected over the ten comparisons ($\alpha = 0.05$). With b (samples where

model A is correct and B is wrong) and c (samples where B is correct and A is wrong), the discordant pairs split as Binomial($b + c$, 0.5) under the null hypothesis of equal error rates. The exact binomial test is used when $b + c < 25$, and the continuity-corrected χ^2 test otherwise.
– Training time and model size (MB) as efficiency metrics.
– Per-script macro F1, recomputed by filtering each model's saved test-set predictions by language_variant, to answer RQ4 without retraining.

5 Results

5.1 RQ1: Five-Way Intent Classification

Experimental Setup. All five models were trained on the 2,472-row training set and evaluated on the 309-row test set using the 5-way category label.
Observations. Table 3 (upper block) shows the full leaderboard. DziriBERT obtains the highest macro F1 (0.9385 single-seed, 95% bootstrap CI [0.9013, 0.9689]; 0.9313 ± 0.0068 across 5 seeds, population SD), followed by the TF-IDF+SVM baseline (0.9145 [0.8733, 0.9498]), MARBERT (0.8952 [0.8547, 0.9307]), AraBERT (0.8871 [0.8441, 0.9211]) and Random Forest (0.7860 [0.7298, 0.8340]). The confidence intervals of the top four models overlap, whereas Random Forest's interval lies entirely below those of the other four. Pairwise McNemar tests with Holm correction show that DziriBERT is significantly better than AraBERT ($p_{\text{Holm}} = 0.006$) and Random Forest ($p_{\text{Holm}} < 0.001$), but not than SVM ($b = 10$, $c = 3$, raw $p = 0.0923$, $p_{\text{Holm}} = 0.327$) or MARBERT ($p_{\text{Holm}} = 0.101$); SVM and MARBERT also significantly outperform Random Forest. The 5-seed MARBERT mean (0.8842) is below AraBERT's (0.8875), so MARBERT and AraBERT are within noise of each other.

Table 3: Leaderboards for five-way intent classification (RQ1) and binary adversarial detection (RQ2). CI: 95% bootstrap on macro F1.

Task	Model	Acc.	Prec.	Rec.	Macro F1 [95% CI]
Five-way	DziriBERT	0.958	0.932	0.946	0.939 [0.901, 0.969]
	SVM	0.935	0.923	0.907	0.914 [0.873, 0.950]
	MARBERT	0.919	0.896	0.906	0.895 [0.855, 0.931]
	AraBERT	0.900	0.866	0.918	0.887 [0.844, 0.921]
	RandomForest	0.838	0.840	0.750	0.786 [0.730, 0.834]
Binary	DziriBERT	0.974	0.974	0.974	0.974 [0.955, 0.990]
	SVM	0.964	0.965	0.964	0.964 [0.942, 0.984]
	MARBERT	0.935	0.935	0.935	0.935 [0.906, 0.961]
	AraBERT	0.929	0.932	0.929	0.929 [0.900, 0.955]
	RandomForest	0.916	0.922	0.916	0.916 [0.883, 0.948]

Per-class results for the model with the highest macro F1 (DziriBERT; Table 4) show near-perfect performance on hate_speech (F1 = 1.000) and safe (F1 = 0.984), but visibly weaker performance on tool_misuse (F1 = 0.831, precision 0.794) and sensitive_request (F1 = 0.919).
Discussion. The dialect-matched encoder (DziriBERT) obtains the highest macro F1, and its advantage over the MSA-centric AraBERT is significant after

Table 4: Per-class results, DziriBERT, intent task.

Class	Prec.	Rec.	F1	n
hate_speech	1.000	1.000	1.000	37
prompt_injection	0.979	0.940	0.959	50
safe	0.993	0.974	0.984	155
sensitive_request	0.895	0.944	0.919	36
tool_misuse	0.794	0.871	0.831	31

Holm correction, consistent with the hypothesis that in-dialect pre-training is beneficial for Algerian-specific data. However, McNemar's test does not find the advantage over the mixed-dialect MARBERT or over the cheaper TF-IDF+SVM baseline statistically significant, which suggests that a strong lexical (character n-gram) baseline is competitive with fine-tuned transformers for this dataset size and task. The classes with the least support (308 and 368 samples in the full dataset for tool_misuse and sensitive_request, respectively) are the weakest ones; 11 of DziriBERT's 13 errors involve tool_misuse (Fig. 3b), mostly confusions with prompt_injection and sensitive_request (only one tool_misuse sample is predicted as safe), which is consistent with its lower precision (0.794).

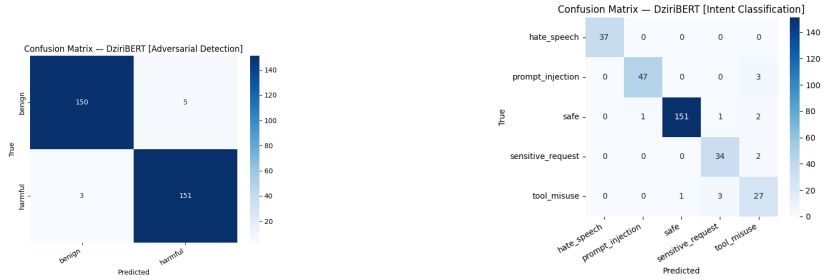
5.2 RQ2: Binary Adversarial Detection

Experimental Setup: The same five models were trained on the binary is_adversarial label with identical splits and preprocessing.

Observations: Table 3 (lower block) shows that all five models achieve substantially higher macro F1 on the binary task than on the 5-way task. DziriBERT again ranks first, followed by SVM, MARBERT, AraBERT and Random Forest. The DziriBERT and SVM intervals overlap (0.955–0.984). The exact binomial McNemar's test between DziriBERT and SVM (the top two) gives $b = 8$, $c = 5$, $p = 0.5811$ – not significant at $\alpha = 0.05$. After Holm correction, DziriBERT is significantly better than AraBERT ($p_{\text{Holm}} = 0.023$) and Random Forest ($p_{\text{Holm}} = 0.013$) but not than MARBERT ($p_{\text{Holm}} = 0.159$); SVM is significantly better than Random Forest ($p_{\text{Holm}} = 0.023$).

DziriBERT's per-class results are near-symmetric (benign: P=0.980, R=0.968, F1=0.974, n = 155; harmful: P=0.968, R=0.981, F1=0.974, n = 154), indicating no marked bias toward either class (5 benign texts flagged, 3 harmful texts missed; Fig. 3a).

Discussion. Combining the four adversarial categories into one label makes the task considerably simpler (Table 3), as binary detection requires only the distinction between safe and any other non-safe pattern, not the distinction between four adversarial sub-types that may share a surface vocabulary (e.g. sensitive_request and hate_speech both sometimes target a named individual). A lightweight binary gate-keeping classifier may therefore be sufficient for coarse moderation, with finer-grained categorization left for flagged content. As with RQ1, the difference between DziriBERT and the runner-up (SVM) is not statistically significant.



(a) Binary detection (b) Five-way intent

Fig. 3: Confusion matrices for DziriBERT on the adversarial-detection and five-way intent tasks.

5.3 RQ3: LLM Zero-Shot / Few-Shot Baseline vs. Fine-Tuned Encoders

Experimental Setup. ALLaM-2-7B and GPT-OSS-120B were tested on the same 309-row test set, zero-shot and few-shot, with the prompt configuration of Section 4 and no gradient-based fine-tuning.

Observations. Table 5 situates the LLM baselines against the fine-tuned-encoder leaderboard. In zero-shot prompting, GPT-OSS-120B is 10.0 (five-way) and 7.2 (binary) macro-F1 points below DziriBERT, and ALLaM-2-7B is 22.8 and 12.3 points below. Adding few-shot examples changes the point estimates by +1.3 (five-way) and +2.7 (binary) points for GPT-OSS-120B and by +5.2 points for ALLaM-2-7B on the binary task, but lowers ALLaM-2-7B on the five-way task by 2.6 points. Every few-shot interval overlaps the corresponding zero-shot interval and no paired test was run, so these changes are suggestive rather than established. With few-shot examples the gap to DziriBERT is 8.8 (five-way) and 4.6 (binary) points for GPT-OSS-120B, and 25.4 and 7.1 points for ALLaM-2-7B. Answer coverage was 100% in every configuration except ALLaM-2-7B zero-shot on the five-way task (99.7%) and few-shot on the binary task (99.0%: refusals on 3 of the 154 adversarial texts). On the five-way task, GPT-OSS-120B is above Random Forest on the point estimate in both settings (zero-shot 0.838, few-shot 0.851, vs. 0.786) and below MARBERT (0.895) and AraBERT (0.887), with overlapping intervals in every case; on the binary task, few-shot GPT-OSS-120B matches AraBERT (0.929) and is above Random Forest (0.916) on the point estimate, again with overlapping intervals.

On the binary task in zero-shot prompting (Fig. 4), the two LLMs err in different directions: GPT-OSS-120B almost never flags benign text (1 of 155 benign texts) but misses 29 of 154 adversarial texts (harmful recall 0.812), whereas ALLaM-2-7B errs in both directions (25 benign texts flagged, 21 adversarial texts missed). With few-shot examples, GPT-OSS-120B still rarely flags benign text (4 of 155, up from 1) and misses fewer adversarial texts (18 of 154; harmful recall 0.883, up from 0.812), while ALLaM-2-7B flags 14 benign texts (down from 25), misses 13 adversarial texts and returns no usable answer for 3 more (counting these refusals as adversarial would raise its macro F1 from 0.903 to 0.913). On the five-way task (Fig. 5), few-shot GPT-OSS-120B is strongest on

Table 5: LLM baselines vs. the two best fine-tuned models, by task (macro F1 [95% bootstrap CI]; LLM rows use strict scoring, i.e. unusable answers count as errors).

Model	Intent Macro F1	Adv. Macro F1
DziriBERT	0.939 [0.901, 0.969]	0.974 [0.955, 0.990]
SVM	0.914 [0.873, 0.950]	0.964 [0.942, 0.984]
GPT-OSS-120B few-shot	0.851 [0.804, 0.896]	0.929 [0.899, 0.955]
GPT-OSS-120B zero-shot	0.838 [0.785, 0.887]	0.902 [0.870, 0.932]
ALLaM-2-7B few-shot	0.684 [0.629, 0.734]	0.903 [0.870, 0.935]
ALLaM-2-7B zero-shot	0.710 [0.653, 0.765]	0.851 [0.809, 0.890]

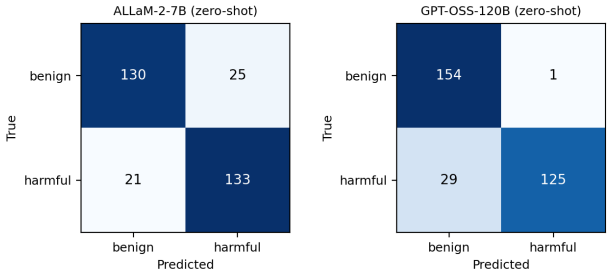


Fig. 4: Zero-shot confusion matrices for the LLM baselines (ALLaM-2-7B, GPT-OSS-120B), binary task.

safe (F1 0.965) and hate_speech (0.946) and weakest on tool_misuse (F1 0.684, precision 0.578), with 10 of the 36 sensitive_request texts labelled tool_misuse (sensitive_request recall 0.694). ALLaM-2-7B labels 20 of the 31 tool_misuse texts as sensitive_request (tool_misuse F1 0.291) and assigns 42 of the 155 safe texts (27.1%) to an adversarial category (safe recall 0.729).

Discussion. In zero-shot prompting the fine-tuned encoders remain ahead of both LLMs, but the gap depends strongly on the LLM: GPT-OSS-120B comes much closer to DziriBERT on the binary task than ALLaM-2-7B does. The error patterns on the binary task do not support a general tendency of LLMs to over-flag benign Algerian Arabic text, although ALLaM-2-7B in the five-way few-shot setting assigns 27.1% of safe texts to an adversarial category. Few-shot examples help most where the zero-shot baseline was weakest (ALLaM-2-7B, binary) but do not close the gap to the fine-tuned encoders, and they did not help ALLaM-2-7B on the five-way task, where tool_misuse is mostly confused with sensitive_request. Because only one example draw was run, the effect of few-shot prompting cannot be separated from the effect of the particular examples drawn. This experiment tests the LLMs as label-outputting classifiers, not their refusal behavior in chat-like interaction (see Section 7).

5.4 RQ4: Per-Script Performance Breakdown

Experimental Setup. For all five models and both tasks, macro F1 was recomputed on each script subset of the saved test-set predictions (Algerian Derja: n = 102; Arabizi: n = 126; Franco-Arabic: n = 81).

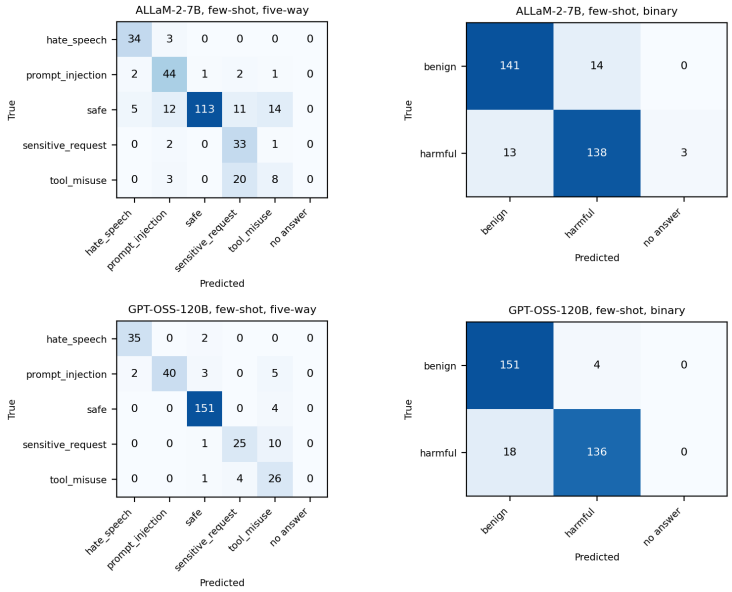


Fig. 5: Few-shot confusion matrices (single seed-42 example draw) for ALLaM-2-7B and GPT-OSS-120B on the five-way intent and binary tasks

Observations: Table 6 shows a consistent pattern across every model on the intent task: Arabizi is the hardest script, with macro F1 dropping 8.3–23.0 points relative to Algerian Derja for every transformer model (MARBERT: 0.986 → 0.756; AraBERT: 0.949 → 0.783; DziriBERT: 0.977 → 0.894). The SVM baseline shows the smallest Arabizi gap (0.914 → 0.889) of any model on the intent task. On the binary task the gap is large for MARBERT (1.000 → 0.839) and AraBERT (1.000 → 0.841), small for DziriBERT (0.990 → 0.960) and negligible for the SVM (0.960 → 0.960); Random Forest does not follow the pattern (0.912 on Derja vs. 0.927 on Arabizi).

Table 6: Per-script macro F1, intent task

Model	Derja	Arabizi	Franco-Ar.
MARBERT	0.986	0.756	0.925
AraBERT	0.949	0.783	0.891
DziriBERT	0.977	0.894	0.942
SVM	0.914	0.889	0.929
RandomForest	0.781	0.693	0.704

Discussion: The Arabizi performance gap is the most prominent script-dependent robustness finding in this paper and is directly relevant to AAAC's safety motivation. The Latin-script, digit-augmented orthography of Arabizi (e.g., 3, 7, 9 for Arabic phonemes) is underrepresented in the pretraining corpora of all three transformer encoders, none of which were pretrained with substantial Arabizi exposure. Even the dialect-matched DziriBERT, which has the smallest gap among the transformers (8.3 points), loses accuracy on Arabizi. The comparatively flat performance of the SVM across scripts is consistent with character

n-gram TF-IDF being less sensitive to script identity than a subword tokenizer trained primarily on Arabic-script vocabulary. A moderation system based on standard Arabic-pretrained encoders may therefore be easier to circumvent by writing adversarial content in Arabizi than in Arabic script. This is based on point estimates only (the SVM gap is just 2.5 points, and no per-script CIs were computed).

5.5 Efficiency Comparison

Table 7 reports training time and model size for all ten configurations. The transformer models take 85–130 seconds (seed-42 run, including model loading) and 498–651 MB, respectively, while the TF-IDF+SVM baseline is trained in less than 0.2 seconds and less than 1 MB. This shows a large efficiency-accuracy trade-off as SVM's macro F1 is only 1.0–2.4 points behind the highest-scoring transformer depending on the task.

Table 7: Training time (seed-42 wall-clock, including model loading) and model size by model and task.

	Intent		Adversarial	
Model	Time (s)	Size (MB)	Time (s)	Size (MB)
MARBERT	129.4	651.4	114.0	651.4
AraBERT	108.6	540.8	84.7	540.8
DziriBERT	90.3	497.8	104.0	497.8
SVM	0.16	0.87	0.03	0.17
RandomForest	2.11	43.9	1.79	27.3

6 Discussion

The four RQs collectively support three main claims. First, dialect-matched pretraining (DziriBERT) achieved the highest observed macro F1 among the evaluated models on both AAAC tasks, significantly ahead of the MSA-centric AraBERT and of Random Forest after Holm correction, but not of TF-IDF+SVM or MARBERT on either task, so any strong claim that dialect-specific pretraining is necessary for strong performance on AAAC is tempered. Second, the fine-tuned encoders remain ahead of the LLMs we tested in both zero-shot and few-shot prompting (RQ3; few-shot results are based on a single example draw), although GPT-OSS-120B is above Random Forest on the five-way task (point estimate).

Third, the per-script analysis (RQ4) shows that aggregate metrics can hide a script-dependent robustness gap: on the five-way task, every classifier scores lower on Arabizi than on Arabic-script Derja (point estimates; no per-script CIs were computed); on the binary task this holds for the three transformers, whereas the SVM is flat and Random Forest does not follow the pattern. This is a potential blind spot that an adversarial actor could exploit.

7 Limitations

- *Single annotator:* Labels were validated by one annotator, so inter-annotator agreement (e.g., Cohen's κ) could not be measured; boundary cases such as safe vs. sensitive_request may therefore be more ambiguous.
- *Small test support:* The 309-row test set has per-class support ranging from 31 to 155, leading to sampling variability. The 95% bootstrap CIs are 6.8–10.4 points wide for the five-way task and 3.6–6.5 points for the binary task (Table 3); no CIs were computed for per-class (Table 4) or per-script (Table 6) scores.
- *Benchmark scope:* The study evaluates supervised classifiers and two LLMs as zero/few-shot classifiers, not the refusal/compliance behavior of deployed conversational LLMs.
- *LLM-generated adversarial data:* Some adversarial examples were generated with Claude and Llama-3.3-70B and manually reviewed (Table 1). This may limit attack diversity, introduce model-specific patterns, and reduce representativeness of real-user attacks.
- *Coverage of Algerian Arabic variation:* AAAC covers three written-script varieties but not regional dialectal variation, Amazigh varieties, register, or speaker demographics, which may affect generalization.

8 Future Work

- Re-annotation of a representative subsample by multiple independent annotators, with inter-annotator agreement (e.g., Cohen's κ) measured and reported.
- Live evaluation of the refusal/compliance behavior of deployed conversational LLMs on AAAC's adversarial prompts.
- Data collection or preprocessing aimed at the performance gap found in RQ4, such as Arabizi-centric pretraining data or script normalization before classification.
- Evaluation of additional LLMs and prompt variants (e.g. per-script few-shot examples and several example draws).
- Independently authored, human-generated adversarial examples, and extension of the two weakest, smallest categories (tool_misuse and sensitive_request).

9 Conclusion

We introduce AAAC, a dataset of 3,090 samples in Algerian Arabic, with five safety-relevant intent classes and three script variants, and evaluate five classifiers and two LLM baselines (ALLaM-2-7B and GPT-OSS-120B) on intent classification and binary adversarial detection. The highest observed macro F1 scores among the evaluated models on both tasks were obtained by a dialect-specific pre-trained encoder (DziriBERT), with macro F1 of 0.939 [0.901, 0.969] and 0.974 [0.955, 0.990] respectively (95% bootstrap CIs); it is not significantly

better than the TF-IDF+SVM baseline or MARBERT at this test-set size (significantly better than AraBERT and Random Forest after Holm correction). The two LLMs trail DziriBERT by 7.2 to 22.8 macro-F1 points in zero-shot prompting and by 4.6 to 25.4 points with few-shot examples (one draw), although GPT-OSS-120B is above Random Forest on the five-way task in both settings (point estimate). On the five-way task, every classifier scores lower on Arabizi than on Arabic-script Derja (point estimates; binary task: see Section 6), a potential robustness gap relevant to AAAC's safety use case.

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